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Understanding Fine-tuning CLIP for Open-vocabulary Semantic Segmentation in Hyperbolic Space

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https://github.com/SJTU-DeepVisionLab/HyperCLIP

Abstract

CLIP, a foundational vision-language model, has emerged as a powerful tool for open-vocabulary semantic segmentation. While freezing the text encoder preserves its powerful embeddings, recent studies show that fine-tuning both the text and image encoders jointly significantly enhances segmentation performance, especially for classes from open sets. In this work, we explain this phenomenon from the perspective of hierarchical alignment, since during fine-tuning, the hierarchy level of image embeddings shifts from image-level to pixel-level. We achieve this by leveraging hyperbolic space, which naturally encoders hierarchical structures. Our key observation is that, during fine-tuning, the hyperbolic radius of CLIP's text embeddings decreases, facilitating better alignment with the pixel-level hierarchical structure of visual data. Building on this insight, we propose HyperCLIP, a novel fine-tuning strategy that adjusts the hyperbolic radius of the text embeddings through scaling transformations. By doing so, HyperCLIP equips CLIP with segmentation capability while introducing only a small number of learnable parameters. Our experiments demonstrate that HyperCLIP achieves state-of-the-art performance on open-vocabulary semantic segmentation tasks across three benchmarks, while fine-tuning only approximately 4% of the total parameters of CLIP. More importantly, we observe that after adjustment, CLIP's text embeddings exhibit a relatively fixed hyperbolic radius across datasets, suggesting that the granularity required for this segmentation task might be quantified using the hyperbolic radius.

1. Introduction

The goal of open-vocabulary semantic segmentation is to develop a segmentation model capable of labeling each pixel in an image with categories that extend beyond a predefined closed set, based on textual descriptions. Vision-language foundation models [7, 12, 13, 15, 22, 23, 26, 28, 32–35, 39, 43, 44, 46, 51, 52, 55, 63], particularly CLIP [46], are frequently employed to provide open-vocabulary recognition capabilities. Consequently, open-vocabulary seman-

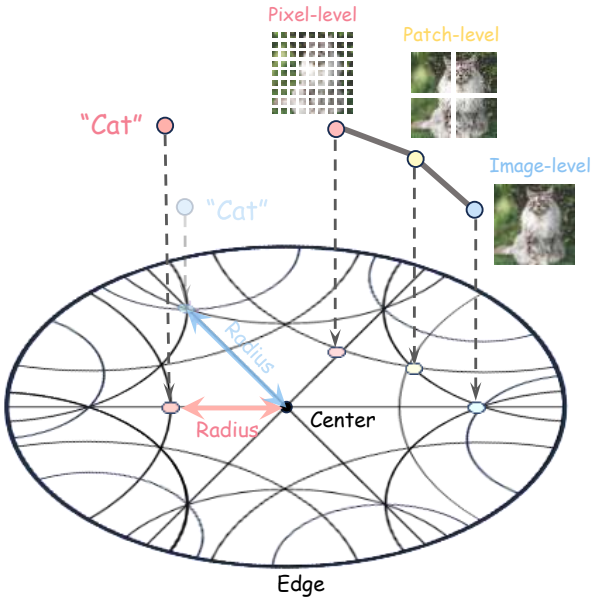


Figure 1. **Visualization of hyperbolic space by a Poincaré ball model**: The height of points reflects the hierarchical level of input data. The black segments refer to the shortest path between the points in the Poincaré ball model. The red arrow line denotes the hyperbolic radius of text embeddings. Embeddings closer to the center have a smaller radius, representing low-level visual information (e.g., pixels), while points farther from the center have a larger radius, representing high-level visual information (e.g., images). During fine-tuning, the hyperbolic radius of text embeddings is adjusted to match the radius associated with pixels, thereby endowing CLIP with segmentation ability (Best viewed in color).

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tic segmentation fundamentally involves transferring these vision-language foundation models, originally trained with image-level supervision, to achieve pixel-level predictions.

Current methods [9, 59, 62, 65] typically fine-tune CLIP on a closed set with segmentation annotations, i.e., COCO [4], to equip it with the segmentation capability. A prevailing opinion is that simply freezing CLIP's text encoder maximally preserves the powerful text embeddings of a vast array of classes, which is believed to be beneficial for good generalization on open sets. Many studies follow this opinion [31, 37, 62]. However, recent studies [9, 42, 59] offer an alternative perspective on fine-tuning: simultaneously fine-tuning both the image and text encoders results in superior segmentation performance on open sets compared to former approaches. To explain and resolve this seemingly paradoxical phenomenon, we first leverage a non-Euclidean manifold—specifically, hyperbolic space [5]—to understand why fine-tuning the text encoder appears to improve segmentation performance on open sets. Building on this understanding, our objective is to design an explainable fine-tuning strategy with a minimal number of introduced tunable parameters, which is able to equip CLIP with strong segmentation performance while minimizing the loss of its generalization ability.

Existing research [1] has found that images inherently exhibit hierarchical structures, consisting of multiple levels: pixels, patches, objects, entire scenes, etc. Given that, in open-vocabulary semantic segmentation, CLIP's image embeddings transition from image level to pixel level, we infer that the performance improvement brought by the recent studies [9, 59] could be attributed to adjusting CLIP's text embeddings from their original hierarchical level to better align with the level represented by pixels, thereby enhancing cross-modal alignment. However, most existing fine-tuning methods operate in Euclidean space, which makes it challenging to quantify such a transition through different hierarchical levels. Hyperbolic spaces have gained significant interest in recent years, owing to their ability to naturally and compactly encode hierarchical structures [47, 48, 57], as shown in Fig. 1. The hyperbolic radius, defined as the distance from a point to the center in hyperbolic space, represents its hierarchical level [50]. Points closer to the center (smaller radius) correspond to low-level visual embeddings, e.g., those from pixels, while points near the edge (larger radius) correspond to high-level visual embeddings, e.g., those from images.

Motivated by this, we project the text embeddings into hyperbolic space and visualize the changes in their radii before and after fine-tuning, as shown in Fig. 2. This fine-tuning is realized using the state-of-the-art method provided in [9]. Specifically, one can observe that the radius after fine-tuning is smaller than that of pre-trained CLIP. In particular, we identify a pattern during fine-tuning: the hierarchical level of the text embeddings adjusts in conjunction with the hierarchical level of the embeddings provided by the image encoder. Our finding brings a novel perspective for fine-tuning CLIP: Adjusting the hyperbolic radius of text embeddings from edge to center endowed CLIP with segmentation ability.

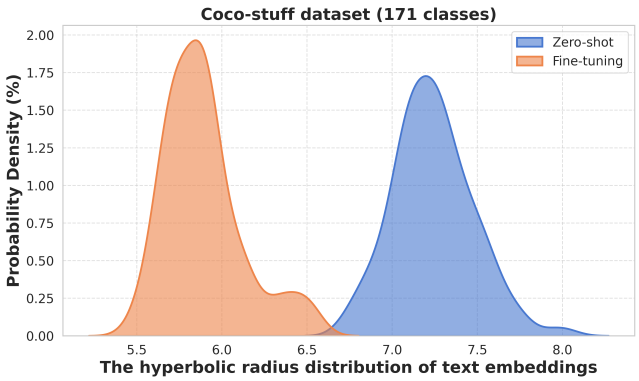


Figure 2. **Understanding fine-tuning CLIP's text encoder from the perspective of hyperbolic radius**. We use hyperbolic radius to illustrate the change in the hierarchical level of 171 classes after fine-tuning. We observe that the hyperbolic radius of each class becomes smaller, suggesting that adjusting the hyperbolic radius equips CLIP with segmentation ability (Best viewed in color).

Building on the analysis above, we propose a novel fine-tuning strategy for CLIP in hyperbolic space, termed HyperCLIP. HyperCLIP aims to directly adjust the hyperbolic radius of CLIP's embeddings to facilitate alignment with the appropriate hierarchical level for segmentation tasks. Specifically, we introduce several block-diagonal scaling matrices. Multiplying these matrices with CLIP's feature representations using a Möbius matrix multiplication operation is equivalent to performing scaling transformations that adjust the radius of CLIP's embeddings. This enables the adjustment of their hierarchical levels, ultimately enhancing the model's performance. As a result, HyperCLIP can equip CLIP with segmentation ability with only a small number of introduced learnable parameters.

Extensive experiments demonstrate that our method sets new state-of-the-art results in open-vocabulary semantic segmentation across three benchmarks, while fine-tuning only approximately 4% of the total parameters of CLIP. More importantly, we observe that after adjustment, CLIP's text embeddings maintain a relatively fixed hyperbolic radius across different datasets. This suggests that the granularity of the segmentation task might be quantified using the hyperbolic radius, warranting further exploration.

2. Related Work

2.1. Open-vocabulary Semantic Segmentation

Prior works on open-vocabulary semantic segmentation typically tackle this task by leveraging CLIP [46]. Early efforts,

e.g., [68], directly fine-tune CLIP on mainstream segmentation datasets, like COCO [4]. However, these approaches argue that fully fine-tuning CLIP's encoder significantly diminishes its ability to generalize to unseen classes. To mitigate this issue, some methods [11, 17, 60, 61] take the opposite approach, fine-tuning an additional mask generator [8] for segmentation while keeping CLIP frozen to preserve its generalization capabilities. However, this approach, with its frozen parameter space, lacks segmentation awareness, leading to a misalignment between regions and text descriptions [37]. Other studies [9, 62, 65] propose more sophisticated solutions that fine-tune only specific parameters, e.g., certain layers of CLIP, enabling pixel-level predictions while keeping the majority of CLIP's parameters fixed to minimize the loss of generalization. While these approaches show significant advantages, they often rely on very small learning rates, implicitly promoting minimal deviations from the pre-trained CLIP, which limits segmentation performance. In summary, there is still limited understanding of how fine-tuning influences CLIP's representations and why fine-tuning CLIP's text encoder appears to enhance segmentation performance. In this paper, we aim to provide several new insights that reveal the underlying logic by incorporating a novel perspective: hyperbolic space.

2.2. Learning in Hyperbolic Spaces

Unlike Euclidean geometry, hyperbolic space can be viewed as the continuous analog of a tree [2], enabling a hierarchical understanding of knowledge structures. Previous research has demonstrated the advantages of representing such hierarchical relationships in various domains, including molecular structures [66], 3D data [6, 56], text data [16, 70], and images [18, 27]. Furthermore, several studies have investigated hyperbolic learning in vision tasks to model hierarchical relationships across different modalities [10, 30, 47], showcasing the benefits of capturing the semantic hierarchy among various data types. Nevertheless, while most studies typically leverage the intrinsic hierarchical properties of hyperbolic space, this paper seeks to capture the detailed changes in semantics reflected by the hyperbolic radius. This approach potentially complements current fine-tuning efforts in open-vocabulary tasks by offering a novel hyperbolic perspective.

3. Preliminary Background

3.1. Hyperbolic geometry

Hyperbolic geometry often has constant negative curvature and is fundamentally different to the Euclidean or spherical geometry which has zero or constant positive curvature, respectively. This enables unique properties such as the divergence of parallel lines and the exponential volume growth towards the boundary. Hyperbolic geometry has five different analytic models to build the hyperbolic space [5]. In this work, we utilize the classical Poincaré ball model as its effectiveness in capturing hierarchical structures [16, 20, 41, 47].

3.2. The Poincaré Ball Model

The Poincaré ball model $(\mathbb{D}_c^n, g^{\mathbb{D}_c})$ with radius $1/\sqrt{c}$ and negative curvature $-c(c > 0)$ is a Riemannian manifold $\mathbb{D}_c^n := \{\mathbf{x} \in \mathbb{R}^n : c\|\mathbf{x}\| < 1\}$ with the metric $g^{\mathbb{D}_c} = \lambda_{c,\mathbf{x}}^2 g^E$, where $\lambda_{c,\mathbf{x}} = \frac{2}{1-c\|\mathbf{x}\|^2}$ is the conformal factor, and $g^E = \mathbf{I}_n$ is the Euclidean metric tensor. The vectorial structure operations defined in Euclidean space can not be directly used in hyperbolic space due to the differential of the space properties. Therefore, we introduce the framework of gyrovector spaces [53, 54] to provide non-associative algebraic operations for hyperbolic space. According to special relativity, when adding speed vectors belonging to the Poincaré ball of radius c, these added speed vectors remain in the ball by the gyrovector spaces operation, also known as the Möbius operation. For instance, the addition and multiplication operations are defined as follows:

Möbius addition operation. For $\mathbf{x}, \mathbf{y} \in \mathbb{D}_c^n$, the Möbius addition operation is defined as:

$$\mathbf{x} \oplus_c \mathbf{y} := \frac{(1 + 2c\langle \mathbf{x}, \mathbf{y} \rangle + c\|\mathbf{y}\|^2)\mathbf{x} + (1 - c\|\mathbf{x}\|^2)\mathbf{y}}{1 + 2c\langle \mathbf{x}, \mathbf{y} \rangle + c^2\|\mathbf{x}\|^2\|\mathbf{y}\|^2}. \quad (1)$$

Möbius scalar multiplication operation. For a scalar $r \in \mathbb{R}$ and a vector $\mathbf{x} \in \mathbb{D}_c^n$, the Möbius scalar multiplication operation is defined as:

$$r \otimes_c \mathbf{x} := (1/\sqrt{c})\tanh(r\tanh^{-1}(\sqrt{c}\|\mathbf{x}\|))\frac{\mathbf{x}}{\|\mathbf{x}\|}. \quad (2)$$

The operation satisfies $r \otimes_c \mathbf{0} := \mathbf{0}$, $r \otimes_c \mathbf{x} = \mathbf{x} \oplus_c \cdots \oplus_c \mathbf{x}$ (r additions), $(r' + r) \otimes_c \mathbf{x} = r \otimes \mathbf{x} \oplus_c r' \otimes_c \mathbf{x}$, $(r'r) \otimes_c \mathbf{x} = r \otimes_c (r' \otimes_c \mathbf{x})$ and $(|r| \otimes_c \mathbf{x})/\|r \otimes_c \mathbf{x}\| = \mathbf{x}/\|\mathbf{x}\|$.

Möbius matrix multiplication operation. Refer to the relation definition in [27], for a matrix $\mathbf{M} \in \mathbb{R}^{n \times n}$ and a vector $\mathbf{x} \in \mathbb{D}_c^n$, if $\mathbf{Mx} \neq \mathbf{0}$, the Möbius matrix multiplication operation is defined as:

$$\mathbf{M} \otimes_c \mathbf{x} := (\frac{1}{\sqrt{c}})\tanh\left(\frac{\|\mathbf{Mx}\|}{\|\mathbf{x}\|}\tanh^{-1}(\sqrt{c}\|\mathbf{x}\|)\right)\frac{\mathbf{Mx}}{\|\mathbf{Mx}\|}.$$

The distance of $\mathbf{x}, \mathbf{y} \in \mathbb{D}_c^n$ is defined as:

$$d_c^{\mathbb{D}}(\mathbf{x}, \mathbf{y}) = (2/\sqrt{c})\tanh^{-1}(\sqrt{c}\| - \mathbf{x} \oplus_c \mathbf{y}\|). \quad (3)$$

The tangent space $\mathcal{T}_{\mathbf{x}}^c\mathbb{D}_c^n$ at a point $\mathbf{x} \in \mathbb{D}_c^n$ is the first order approximation of $\mathbb{D}_c^n$, which is an n-dimensional Euclidean space. The tangent space $\mathcal{T}_{\mathbf{x}}^c\mathbb{D}_c^n$ and $\mathbb{D}_c^n$ are mapped to each other by exponential $(\mathcal{T}_{\mathbf{x}}^c\mathbb{D}_c^n \mapsto \mathbb{D}_c^n : \exp_{\mathbf{x}}^{\mathbb{D},c}(\cdot))$ and logarithmic $(\mathbb{D}_c^n \mapsto \mathcal{T}_{\mathbf{x}}^c\mathbb{D}_c^n : \log_{\mathbf{x}}^{\mathbb{D},c}(\cdot))$ maps, respectively. For any $\mathbf{x}, \mathbf{y} \in \mathbb{D}_c^n$ and $\mathbf{v} \in \mathcal{T}_{\mathbf{x}}^c\mathbb{D}_c^n$, the mapping functions

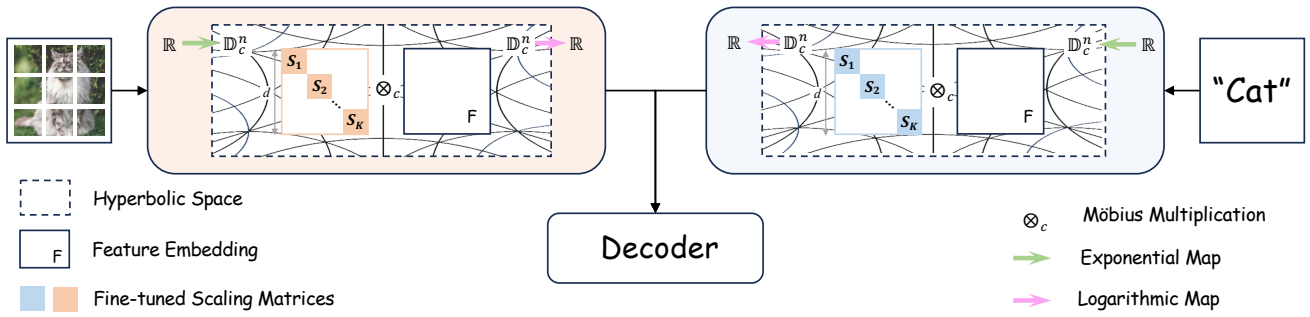


Figure 3. **A schematic representation of HyperCLIP.** In the HyperCLIP framework, we fine-tune CLIP in hyperbolic space by minimizing its embeddings' hyperbolic radii using several tunable block-diagonal scaling matrices.

are given for $\mathbf{v} \neq \mathbf{0}$ and $\mathbf{y} \neq \mathbf{x}$ by:

$$\exp_{\mathbf{x}}^{\mathbb{D},c}(\mathbf{v}) = \mathbf{x} \oplus_c \left(\tanh\left(\sqrt{c} \frac{\lambda_{c,\mathbf{x}} \|\mathbf{v}\|}{2} \right) \frac{\mathbf{v}}{\sqrt{c}\|\mathbf{v}\|} \right), \quad (4)$$

$$\log_{\mathbf{x}}^{\mathbb{D},c}(\mathbf{y}) = \frac{2}{\sqrt{c}\lambda_{c,\mathbf{x}}} \tanh^{-1}\left(\sqrt{c} \| -\mathbf{x} \oplus_c \mathbf{y} \| \right) \frac{-\mathbf{x} \oplus_c \mathbf{y}}{\| -\mathbf{x} \oplus_c \mathbf{y} \|}. \quad (5)$$

4. Methodology

In this section, we introduce HyperCLIP for open-vocabulary semantic segmentation. HyperCLIP is based on the Poincaré ball model, which aims to adjust the hyperbolic radius of CLIP's embeddings, thereby equipping CLIP with segmentation ability.

4.1. Fine-tuning in Hyperbolic Space

The core technique in HyperCLIP involves linear transformations, i.e., scaling, in the hyperbolic space. One of the primary methods [64] to learn the weights of linear transformations in hyperbolic space is hyperLoRA [64]. Given a feature embedding $\mathbf{z}$ in Euclidean space, this method first maps $\mathbf{z}$ to the hyperbolic space at a local reference point, typically the center, using the exponential map (e.g., Eq. 4). Linear transformations with learnable weight matrix $\mathbf{W}$ is then applied to $\mathbf{z}$ within the hyperbolic space to obtain the hyperbolic representation. Then, a logarithmic map (e.g., Eq. 5) is applied to map the hyperbolic representation back to the Euclidean space. Formally, in the Poincaré ball model, this process is realized as:

$$\mathbf{W}\mathbf{z} = \log_{\mathbf{0}}^{\mathbb{D},c}(\mathbf{W} \otimes_c \exp_{\mathbf{0}}^{\mathbb{D},c}(\mathbf{z})), \quad (6)$$

where $\otimes_c$ is the möbius matrix multiplication operation. The primary aim of this method [64] is to capture more intricate hierarchical relationships within hyperbolic space. In contrast, our goal is to directly adjust the hyperbolic radius of CLIP's embeddings, leading to a scaling transformation.

4.2. Hyperbolic Radius Adjustment

The objective of HyperCLIP is to introduce a scaling transformation to directly adjust the hyperbolic radii of CLIP's embeddings in hyperbolic space, thereby equipping CLIP with segmentation ability. Concretely, it necessitates the utilization of a diagonal matrix, denoted as $\mathbf{S}$, to scale a CLIP's feature embedding $\mathbf{z} \in \mathbb{R}^{b \times d}$, where d is the feature dimension of $\mathbf{z}$, b is the spatial dimension of input images or number of classes. When extending this process to hyperbolic space, $\mathbf{S}$ can directly adjust the hyperbolic radius of $\mathbf{z}$ (this deviation can be found in Sec. 4.3). Specifically, we realize the diagonal matrix $\mathbf{S}$ in a block-wise manner,

$$\mathbf{S} = \mathrm{diag}(\mathbf{S}_1, \mathbf{S}_2, \ldots, \mathbf{S}_K), \quad (7)$$

where $\mathbf{S}_k \in \mathbb{R}^{n \times n}$ denotes a square matrix of the k-th block, and the block number K is calculated by $K = d/n$. By adjusting the size of n, we can achieve a transition between naive scaling transformation ($n = 1$, i.e., modifying only the diagonal elements) and arbitrary transformation ($n = d$, i.e., fully fine-tuning in hyperbolic space). The ablation study of n is presented in Sec. 5.3.

Overall Fine-tuning. During the fine-tuning phase, HyperCLIP introduces a similar process to Eq. 6 is incorporated into CLIP's encoder. Specifically, HyperCLIP uses the following forward pass:

$$\mathbf{S}\mathbf{z} = \log_{\mathbf{0}}^{\mathbb{D},c}(\mathbf{S} \otimes_c \exp_{\mathbf{0}}^{\mathbb{D},c}(\mathbf{z})). \quad (8)$$

Concurrently, the original components of CLIP load their weights from the pre-trained checkpoint, with their parameters remaining frozen.

Parameter complexity. For an K-block diagonal matrix, the number of parameters is d^2/K, resulting in a complexity of $\mathcal{O}\left(d^2/K\right)$. We can optionally share the block matrix to reduce the number of parameters, i.e., $\mathbf{S}_k = \mathbf{S}_j, \forall k \neq j$. This reduces the parameter complexity to $\mathcal{O}\left(d^2/K^2\right)$. We can further reduce parameter complexity by setting $n = 1$, which transforms the block-diagonal matrix into a diagonal

Model	VLM	Additional Backbone	Fine-tuning Space	A-847	PC-459	A-150	PC-59	PAS-20	PAS-20^b
Partial Fine-Tuning									
ZS3Net [3]	-	ResNet-101	E	-	-	-	19.4	38.3	-
LSeg [31]	CLIP ViT-B/32	ResNet-101	E	-	-	-	-	47.4	-
OpenSeg [17]	ALIGN	ResNet-101	E	4.4	7.9	17.5	40.1	-	63.8
ZegFormer [11]	CLIP ViT-B/16	ResNet-101	E	4.9	9.1	16.9	42.8	86.2	62.7
ZSseg [61]	CLIP ViT-B/16	ResNet-101	E	7.0	-	20.5	47.7	88.4	-
OVSeg [37]	CLIP ViT-B/16	ResNet-101c	E	7.1	11.0	24.8	53.3	92.6	-
ZegCLIP [69]	CLIP ViT-B/16	-	E	-	-	-	41.2	93.6	-
Selective Fine-Tuning									
SED [59]	CLIP ConvNeXt-B	-	E	11.4	18.6	31.6	57.3	94.4	-
CAT-Seg [9]	CLIP ViT-B/16	-	E	<u>12.0</u>	<u>19.0</u>	<u>31.8</u>	<u>57.5</u>	<u>94.6</u>	<u>77.3</u>
Parameter-efficient Fine-Tuning									
SAN [62]	CLIP ViT-B/16	Side Adapter	E	10.1	12.6	27.5	53.8	94.0	-
Ours	CLIP ViT-B/16	-	**H**	**12.3**	**19.2**	**32.1**	**58.5**	**95.6**	**78.9**
Partial Fine-Tuning									
LSeg [31]	CLIP ViT-B/32	ViT-L/16	E	-	-	-	-	52.3	-
OpenSeg [17]	ALIGN	Eff-B7	E	8.1	11.5	26.4	44.8	-	70.2
OVSeg [37]	CLIP ViT-L/14	Swin-B	E	9.0	12.4	29.6	55.7	94.5	-
SAN [62]	CLIP ViT-L/14	-	E	12.4	15.7	32.1	57.7	94.6	-
ODISE [60]	CLIP ViT-L/14	Stable Diffusion	E	11.1	14.5	29.9	57.3	-	-
FC-CLIP [65]	CLIP ConvNeXt-L	-	E	14.8	18.2	34.1	58.4	95.4	-
Selective Fine-Tuning									
SED [59]	CLIP ConvNeXt-L	-	E	13.9	22.6	35.2	60.6	96.1	-
CAT-Seg [9]	CLIP ViT-L/14	-	E	<u>16.0</u>	<u>23.8</u>	<u>37.9</u>	<u>63.3</u>	<u>97.0</u>	<u>82.5</u>
Parameter-efficient Fine-Tuning									
SAN [62]	CLIP ViT-L/14	Side Adapter	E	12.4	15.7	32.1	57.7	94.6	-
Ours	CLIP ViT-L/14	-	**H**	**16.3**	**24.1**	**38.2**	**64.2**	**98.3**	**84.3**

Table 1. **Comparison with state-of-the-art methods on standard benchmarks.** The best-performing results are presented in bold, while the second-best results are underlined. "E": Euclidean Space. "H": Hyperbolic Space.

matrix. In this case, the number of parameters becomes d, resulting in a complexity of $\mathcal{O}(d)$. Despite all these strategies to improve parameter efficiency, we note that the resulting matrix $\mathbf{S}$ remains to realize scaling transformations, thereby allowing it to adjust the radius of CLIP's embeddings.

4.3. Theoretical Analysis

To further demonstrate the effectiveness of the proposed HyperCLIP method, we conduct a theoretical analysis of the scaling transformation. Theorem 1 reveals that the employed scaling matrix can directly adjust the hyperbolic radius, thus enabling transferring through different hierarchical levels. Detailed proof refers to the supplementary.

Theorem 1 (Scaling). According to the definitions of the classical Poincaré Ball model and the hyperbolic tangent, for a point $\mathbf{x} \in \mathbb{D}_c^n$ in hyperbolic space, its hyperbolic radius is defined as:

$$\text{Rad}_{\mathbf{x}} := d_c^{\mathbb{D}}(\mathbf{x}, \mathbf{0}) = (\frac{2}{\sqrt{c}})\tanh^{-1}(\sqrt{c}\|\mathbf{x}\|), \tag{9}$$

where $\mathbf{0}$ is the center of the hyperbolic space. Let s is a scaling parameter, combine with Eq. 2 and Eq. 9, the radius of the point $\mathbf{x}_s = \mathcal{T}_{\mathbf{x}}(s) := s \otimes_c \mathbf{x}$ which changed by the scaling transformation $\mathcal{T}_{\mathbf{x}}(s)$ is obtained by:

$$\begin{aligned}\text{Rad}_{\mathbf{x},s}(s) &= \frac{2}{\sqrt{c}}\tanh^{-1}(\sqrt{c}\|\mathbf{x}_s\|) \\ &= \frac{2}{\sqrt{c}}\tanh^{-1}(\sqrt{c}\left\|\frac{\tanh(\frac{s\sqrt{c}}{2}\text{Rad}_{\mathbf{x}})\mathbf{x}}{\sqrt{c}\|\mathbf{x}\|}\right\|).\end{aligned} \tag{10}$$

It is noticed that the $\text{Rad}_{\mathbf{x},s}(s)$ satisfies the monotonicity criteria, therefore, we can adjust the hyperbolic radius by changing the scaling parameter s. Refer to the definition of the Möbius matrix multiplication operation in [27], the scaling matrix $\mathbf{S}$ also has the ability to adjust the hyperbolic radius via the scaling transformation $\mathcal{T}_{\mathbf{x}}(\mathbf{S})$. ■

5. Experiments

5.1. Experimental Setup

Datasets. Following prior works [9, 59], we use the COCO-Stuff dataset [4] as our training set. This dataset consists of around 118,000 densely annotated images, covering 171 distinct semantic categories. For inference, we compare our

Methods	Space for Fine-tuning	AP_{50}^{Base}	AP_{50}^{Novel}
CLIP	-	21.6	36.4
Other Fine-Tuning			
CLIM [58]	E	25.7	42.5
CAT-Seg [9]	E	25.2	42.3
Parameter-efficient Fine-Tuning			
LoRA [21]	E	24.4	41.5
HyperLoRA [64]	H	25.1	42.0
Ours	H	**26.3**	**42.7**

Table 2. **Comparisons on open-vocabulary object detection task** on the COCO [38] dataset. Results are measured by the box AP at IoU threshold 0.5. "E": Euclidean space. "H": hyperbolic space.

method against state-of-the-art approaches across several semantic segmentation benchmarks, including ADE20K [67], PASCAL VOC [14], and PASCAL-Context [40].

- **ADE20K** [67] is a well-established semantic segmentation dataset with approximately 20,000 training images and 2,000 validation images. It also includes two separate test sets: `A-150`, consisting of 150 common categories, and `A-847`, which contains 847 categories.
- **PASCAL VOC** [14] is a smaller dataset for semantic segmentation, consisting of 1464 training images and 1449 validation images, covering 20 different foreground categories. We refer to this dataset as `PAS-20`. In line with [9], we also report scores on `PAS-20`^{b}, which includes the "background" as the 21st category.
- **PASCAL-Context** [40] is an extended version of the original PASCAL VOC dataset. It provides two test sets: `PC-59` and `PC-459`, containing 59 and 459 categories respectively, for evaluation purposes.

Evaluation Metric. Consistent with prior studies [9, 59], we use mean Intersection over Union (mIoU) to evaluate the semantic segmentation performance on these benchmarks.

Implementation Details. We implement our method using the Transformer-based CLIP model. Following the protocol established in [9], we evaluate our results on two versions of the CLIP model: ViT-B/16 and ViT-L/14. For training, we follow the same experimental setup as [9]. Besides, we use the Adam optimizer [29] with an initial learning rate of 1×10^{-5} for CLIP, and a weight decay of 10^{-4}. Training is conducted with one image per mini-batch. We apply the block-diagonal learnable weight in the self-attention module in each layer of CLIP. We employ the cost-based approach from [9] as our decoder. The curvature c of Poincaré ball model is set to 0.01. All models are trained for 80,000 iterations using 4 NVIDIA A800 GPUs.

5.2. Comparisons with State-of-the-art Methods

Here, we compare our proposed method with several state-of-the-art methods, as shown in Table 1, using six test sets

Method	ODISE [60]	FC-CLIP [65]	MAFT+ [25]	Ours
PQ	23.4	26.8	27.1	**29.2**

Table 3. **Comparisons on open-vocabulary panoptic segmentation task** on the ADE20K [67] dataset. "PQ": Panoptic quality.

n	Param. (M)	`A-847`	`PC-459`	`A-150`	`PC-59`	`PAS-20`^{b}
1	0.04	5.9	9.4	27.4	52.4	74.9
8	0.35	10.1	15.4	29.9	54.2	75.6
16	0.70	11.4	17.4	31.0	55.9	76.2
32	1.41	11.7	18.1	31.7	56.6	76.7
64	2.81	11.9	18.7	32.0	57.8	77.8
128	5.63	**12.3**	**19.2**	**32.1**	<u>58.5</u>	<u>78.9</u>
256	11.25	<u>12.2</u>	<u>19.0</u>	<u>31.9</u>	**58.8**	**79.3**

Table 4. **Ablation on the size of diagonal blocks n.** The base model is ViT-B/16. The best-performing results are presented in bold, while the second-best results are underlined.

Methods	OVSeg [37]	CAT-Seg [9]	SAN [62]	Ours
Param. (M)	147.2	39.5	8.4	**5.6**

Table 5. **Efficiency comparison** in terms of learnable parameters for fine-tuning CLIP. The base model is ViT-B/16.

across three benchmarks. Overall, we achieve the best results. Most existing open-vocabulary semantic segmentation methods follow partial fine-tuning strategies, i.e., fine-tuning CLIP's image encoder. Although these methods provide sufficient flexibility for aligning with text descriptions generated by the text encoder, they fail to adjust the text embeddings to the appropriate hierarchical level for effective text-to-image alignment, often leading to suboptimal segmentation performance. Differently, CAT-Seg [9] simultaneously fine-tunes both the text encoder and image encoder of CLIP, achieving performance comparable to ours on some of the datasets. However, its fine-tuning scheme is manually controlled through different layer combinations, necessitating a careful design to balance generalization and segmentation ability, while ours does not suffer from such an issue. Then, compared to SAN [62], another parameter-efficient fine-tuning method that introduces only a limited number of tunable parameters, our approach significantly outperforms it, achieving improvements of 6.6% on the `PC-459` dataset and 4.7% on the `PC-59` dataset with ViT-B/16 as the base model. These results demonstrate the effectiveness of our method in preserving generalization while mastering segmentation capability.

Efficiency comparison. We compare the efficiency of our method with other approaches, including OVSeg [37], CAT-Seg [9], and SAN [62], all of which utilize a ViT version of CLIP. The comparison, summarized in Table 5, shows

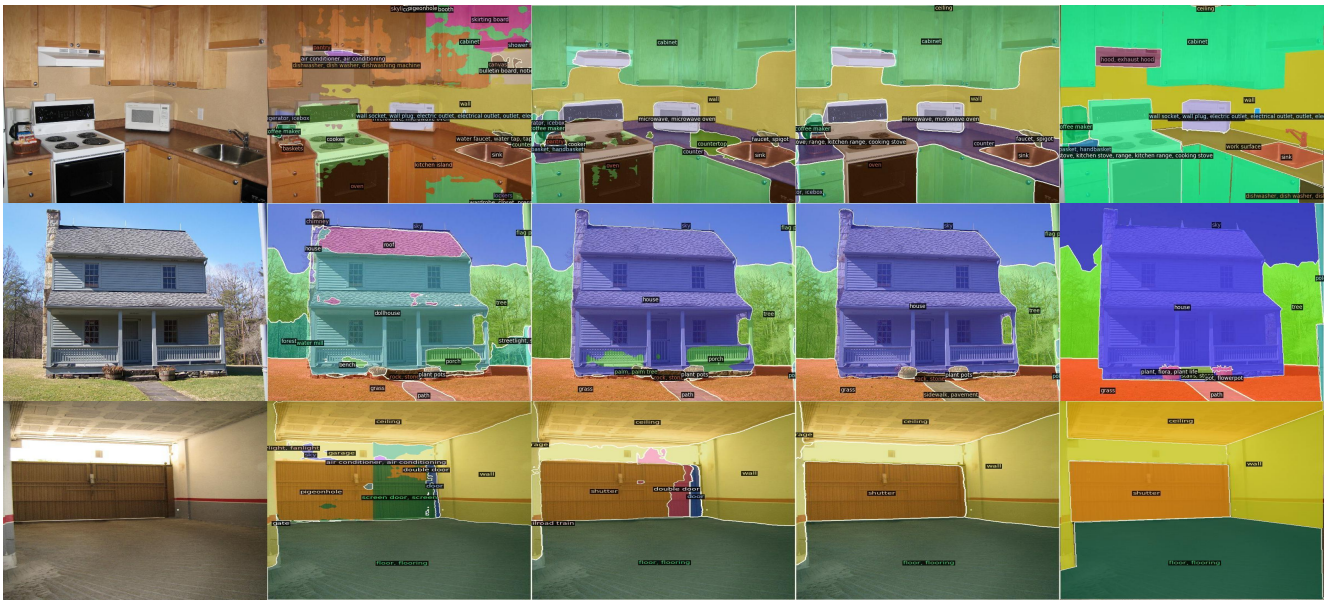


Image SAN [62] CAT-Seg [9] Ours Ground Truth

Figure 4. Comparison of qualitative reults on ADE20K [67] with 847 categories. We compare our method with other two state-of-the-art methods, i.e., SAN [62] and CAT-Seg [9]. The results show our method performs more precise segmentation in different scenarios.

Method	Image encoder	Text encoder	Param. (M)	A-847	PC-459	A-150	PC-59	PAS-20	PAS-20^b
Freeze	✗	✗	0	4.4	6.6	24.8	49.4	92.5	71.9
Parameter-efficient Fine-Tuning									
LoRA [21]	✓	✓	7.5	11.4	17.6	28.6	55.1	94.2	76.7
OFT [45]	✓	✓	3.8	10.9	18.0	30.2	53.7	93.7	74.3
VPT [24]	✓	✓	2.2	5.7	10.2	23.7	54.3	93.8	75.1
Adapter [19]	✓	✓	7.5	10.4	16.5	28.8	54.9	94.2	75.2
SSF [36]	✓	✓	0.6	6.9	15.2	28.6	52.1	93.2	72.8
HyperLoRA [64]	✓	✓	7.5	11.4	18.2	29.7	55.8	94.3	77.2
HyperCLIP (Ours)	✓	✗	3.4	10.3	16.8	30.5	57.6	94.4	78.0
	✗	✓	2.2	11.1	17.5	29.8	55.1	93.7	76.1
	✓	✓	5.6	**12.3**	**19.2**	**32.1**	**58.5**	**95.6**	**78.9**

Table 6. **Ablation study on the fine-tuning encoder in HyperCLIP and comparison of universal PEFT methods**, including "LoRA" [21], "OFT" [45], "VPT" [24], "Adapter" [19], "LST" [49], "SSF" [36] and "HyperLoRA" [64] which is a recent PEFT method fine-tunes large language models in hyperbolic space. For a clear comparison, we ignore the segmentation decoder when calculating the number of learnable parameters. The base model is ViT-B/16. ✓: Fine-tuning. ✗: Freezing.

that our method uses the fewest trainable parameters while equipping CLIP with segmentation ability. Since we do not introduce any extra architectures, and the logarithmic and exponential maps can be completed within linear complexity [64], the inference overhead is negligible.

Application to other tasks. Here, we further carry out extra experiments in two other tasks. In an open-vocabulary object detection task, CLIP's image embeddings are transitioned from the image level to the object level. We follow a recent work [58] to investigate open-vocabulary object detection on the COCO dataset [38]. As shown in Table 2, the superior results demonstrate that HyperCLIP can effectively generalize to other tasks that require aligning text embeddings with embeddings from arbitrary hierarchical levels, e.g., object-level alignment. Furthermore, the open-vocabulary panoptic segmentation results presented in Table 3 further corroborate this conclusion.

Qualitative results. Here, we visualize our method's representative example segmentation results against prevailing methods, i.e., CAT-Seg [9] and SAN [62], in the A-847

dataset. As shown in Fig. 4, we can observe that our method can make more accurate predictions on object location and category. Even if objects are small and the background is complicated, our method still performs well.

5.3. Ablation Study

Ablation of fine-tuning CLIP. We assess the efficacy of various fine-tuning design choices for HyperCLIP. As shown in Table 6, we freeze one of CLIP's encoders and compare the performance with several commonly-used PEFT methods, e.g., LoRA [21]. Compared to fully freezing CLIP's encoders (row 1), fine-tuning either encoder independently allows for minimizing its hyperbolic radius, thereby equipping the corresponding encoder with segmentation ability. However, one modality's hierarchical level remains at the image level, which hinders CLIP from fully acquiring segmentation ability. Consequently, jointly fine-tuning both encoders aligns the hierarchical levels at the pixel level, resulting in optimal performance. We also compare our method with existing PEFT strategies. Most of these methods, which operate in Euclidean space, struggle to capture the transition between different hierarchical levels. Notably, HyperLoRA [64] also updates parameters in hyperbolic space. However, because HyperLoRA does not specifically adjust the hyperbolic radius, it fails to fully endow CLIP with segmentation ability, resulting in inferior performance compared to our method.

Ablation of the size of diagonal blocks. The size of the diagonal block in our scaling transformations determines the number of parameters required to equip CLIP with segmentation ability. To investigate its impact, we conduct an ablation study on the scaling transformations by systematically varying the size of the diagonal block. As shown in Table 4, accuracy improves incrementally on datasets with fewer classes. However, on datasets with more classes (e.g., A-847), the accuracy performance reaches a saturation point around a block size of 128. This suggests that increasing the dimensionality of the diagonal block introduces more trainable parameters, which may lead to over-fitting on a closed set.

Effect of the hyperbolic radius of text embeddings. In order to closely examine the effect of the hyperbolic radius on equipping CLIP with segmentation ability, we apply a mean squared error (MSE) loss during fine-tuning. This MSE loss encourages the average hyperbolic radius of the text embeddings to converge towards different target values. As shown in Fig. 5, the initial value of the average hyperbolic radius is approximately 7.5, which is subsequently minimized to around 6.0 to achieve the optimal value for segmentation. Deviating from this value, either by increasing or decreasing the expected radius, results in diminished performance. This observation offers a new insight: the hyperbolic radius required for a vision task might be relatively fixed, suggesting that it could serve as a metric to quantify the granularity of different vision tasks.

Möbius.	A-847	PC-459	A-150	PC-59	PAS-20^b
✗	11.1	18.2	30.2	57.3	77.5
✓	**12.3**	**19.2**	**32.1**	**58.5**	**78.9**

Table 7. **Ablation on the necessity of Möbius matrix multiplication.** "Möbius.": Möbius matrix multiplication.

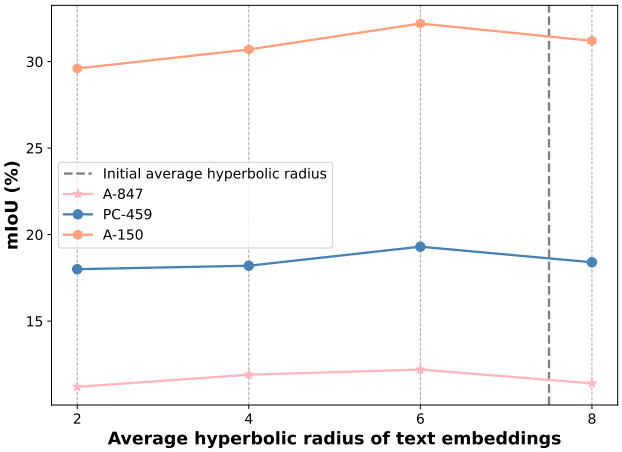


Figure 5. **Visualization on the effect of different average hyperbolic radii of text embeddings.** The controlling of different hyperbolic radii is achieved via a mean squared error (MSE) loss that enforces a constraint between the expected radius and the mean radius of the text embeddings during fine-tuning.

Necessity of Möbius matrix multiplication. We ablate this experiment to validate the influence of the Möbius matrix multiplication operation. In Table 7, "✗" means that we use standard matrix multiplication. It achieves worse performance compared with "✓", which indicates that the Möbius matrix multiplication is essential for minimizing the hyperbolic radius of CLIP's embeddings.

6. Conclusion

In this paper, we propose HyperCLIP, a novel fine-tuning strategy for CLIP tailored to open-vocabulary semantic segmentation. Recognizing the hierarchical structure of images, we hypothesize that aligning the hierarchical level of CLIP's text encoder with pixel-level representations is crucial for accurate pixel-level predictions. HyperCLIP achieves this by adjusting the hierarchical level in hyperbolic space through scaling transformations applied to CLIP's embeddings, enabling precise cross-modal alignment. Extensive experiments demonstrate that HyperCLIP sets new state-of-the-art results in open-vocabulary semantic segmentation. More importantly, we observe that after adjustment, CLIP's text embeddings maintain a relatively fixed hyperbolic radius across datasets. This suggests that the segmentation task's granularity may be quantified using the hyperbolic radius, offering a promising direction for future research.

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